

Yassin Soliman

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SUMMARY

Software Engineering student and GPU library performance engineer. Ships **C++** convolution kernels, solver dispatch, and correctness gates to a production deep-learning library for datacenter GPUs, with **merged upstream contributions** across **CUDA/cuDNN** and **ROCm**.

EDUCATION

University of Calgary, Schulich School of Engineering
Bachelor of Science in Software Engineering

Calgary, AB
Expected April 2028

EXPERIENCE

Software Engineer Intern, AI GPU Software (AGS) Libraries May 2026 – Present
AMD |  |  | **C++, Python, ROCm, MIOpen, hipDNN, SLURM** *Calgary, AB*

- Shipped a **backward-data convolution solver** and 7×7 depthwise dispatch upstream to a production DL library (**MI355X**), reusing the forward kernel with a **rotated filter** (no new kernel or workspace); **ConvNeXt-Large fp16 training +50%**, forward **+67%**
- Found and gated a **silent-correctness defect** in three assembly conv solvers indexing memory with **32-bit offsets** past **INT_MAX**; wrong forward results **6/14** $\rightarrow$ **0/14**, **1.12 $\times$** faster
- Tuned and shipped pre-tuned **kernel-selection databases** for **5 GPU architectures** (95K find / 82K perf entries), with benchmark-confirmed solver overrides up to **3 $\times$** faster
- Cut a **225,990-command** GPU validation suite **98.2%** to 4,096 at full feature coverage (**~ 50 GPU-days** $\rightarrow$ **~ 1**); rebuilt an internal dashboard query path for **483 s** $\rightarrow$ **11 s** at **264M rows**
- Prototyped **hipDNN** heuristic plugins for **SDPA/attention** and convolution regime classification, with engine-ID probing and a MIOpen provider verifier to steer shapes toward faster solvers

Software Lead Nov. 2024 – Present
Waybionic (Student Design Team) |  |  | **ROS 2, C++, Python, RViz, Arduino** *Calgary, AB*

- Co-lead the **ROS 2 (Jazzy)** ground station for a **robotic surgical arm** in **C++/Python**: URDF/Xacro description, **RViz** visualization, diagnostics, telemetry, and safety monitoring; added Apple Silicon development via **RoboStack** with setup, build, and launch scripts
- Direct **10 engineers**, authoring the contributing guide and workflow conventions and reviewing the ROS 2 stack (**19 reviews across 9 PRs**) spanning diagnostics, IMU sensor packages, URDF import, and build automation

OPEN SOURCE CONTRIBUTIONS

Spicetify Marketplace & CLI Contributor |  | **TypeScript, React, IndexedDB, Dexie, Zod** 2026

- Merged **6 upstream PRs**, fixing install-state race conditions with serialized/atomic **IndexedDB** persistence, centralizing **Zod** validation and cache recovery, and hardening locale, color, and installed-state behavior

PROJECTS

AetherVSR (In Development) |  | **TypeScript, WebGPU, WGSL, ONNX Runtime** 2026 – Present

- Shipped a **6,291-parameter neural upscaler** into a **zero-readback WebGPU** pipeline at **6.3 ms p50** (39% of a 60 Hz frame budget) for $1280\times 720 \rightarrow 2560\times 1440$, with automatic fallback to a conventional scaler; measured **247 GMAC/s** fp16 on 3×3 convolution
- Took every GPU figure from **timestamp-query** spans rather than wall-clock subtraction ($n=240$ trailing samples), and tuned tile size, accumulator count, and occupancy against measured throughput

Soundalike |  |  | **Python, PyTorch, CUDA, cuDNN** 2026

- Trained self-supervised audio embeddings on **106K tracks** on a local **RTX 5080**, serving a shipped **272,853-track** catalog; raised training GPU utilization **9%** $\rightarrow$ **99%** and measured a **4.2 $\times$** FP16/channels-last speedup with a custom **cuDNN solver inspector**

WebStrafe |  |  | **TypeScript, Three.js, WebSockets** 2026

- Built a Source-style **128 Hz fixed-timestep** kinematic movement engine with surf/air/ground handling, ramp clipping, **BVH collision**, movement tests, and real-time WebSocket multiplayer

TECHNICAL SKILLS

Languages: C++, C, Python, Java, Go, SQL, TypeScript, JavaScript
GPU / Systems: CUDA, cuDNN, ROCm, MIOpen, hipDNN, WebGPU, WGSL, GPU profiling, Linux, SLURM, Shell/Bash
ML / Computer Vision: PyTorch, NumPy, OpenCV, YOLOv8, ByteTrack, tensor cores, mixed precision
Cloud / DevOps / Web: Docker, Kubernetes, AWS, GitLab, Jenkins, Ansible, CI/CD, microservices, React, Vue, Node.js, React Native, Fastify, Supabase